

Key Matrices

1. Coefficient Matrix A:

The **coefficient matrix** is simply the matrix that represents the coefficients of the variables in a system of linear equations. For example, if the system of equations is:

$$\left. \begin{aligned} x + 2y + 3z &= 4 \\ 4x + 5y + 6z &= 7 \\ 7x + 8y + 9z &= 10 \end{aligned} \right\} \quad A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

2. Upper Triangular Matrix U:

An **upper triangular matrix** is a square matrix where all elements below the main diagonal are zero. The general form of an upper triangular matrix U is:

$$U = \begin{bmatrix} 1 & 4 & 5 \\ 0 & 2 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

3. Lower Triangular Matrix L:

A **lower triangular matrix** is a square matrix where all elements above the main diagonal are zero. The general form of a lower triangular matrix L is:

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 5 & 0 \\ 3 & 7 & 6 \end{bmatrix}$$

4. Elimination Matrix E:

An **elimination matrix** is used in Gaussian elimination to zero out elements of a matrix. For example, if you want to subtract a multiple of one row from another to eliminate a variable, the elimination matrix E represents this operation.

5. Overall Elimination Matrix E:

The **overall elimination matrix** is the product of all elimination matrices used to perform Gaussian elimination. If several elimination steps are needed, the total elimination matrix E is the product of each individual elimination matrix.

6. Inverse Matrix A^{-1}

The **inverse of a matrix** A^{-1} is the matrix such that: $AA^{-1} = I$

where I is the identity matrix. Not all matrices have inverses, and a matrix must be square and have a non-zero determinant to be invertible.

7. Permutation Matrix P :

A **permutation matrix** is a square matrix obtained by permuting the rows or columns of an identity matrix. It is used in row exchanges during Gaussian elimination.

For example, a permutation matrix that swaps the first and second rows of a 3×3 identity matrix is:

$$P = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

8. Transpose Matrix A^T :

The **transpose of a matrix** A , denoted A^T , is formed by flipping the matrix over its diagonal, turning rows into columns and vice versa. For example:

$$A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix} \rightarrow A^T = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

9. Symmetric Matrix S :

A matrix is **symmetric** if it is equal to its transpose, i.e., $S = S^T$. For example:

$$S = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} \quad \text{Here, } S = S^T$$

For $Ax = B$ $\rightarrow$
in A ,

- ⊗ if all of them are independent columns $\rightarrow$ one soln exists
- ⊗ if there are one independent columns $\rightarrow$ no soln exists.
- ⊗ if there are no independent columns $\rightarrow$ infinite soln exists.

Back Substitution to solve $Ox = c$

↳ Explained in writing system in 1.3 part 1.

Gaussian Elimination

⊗ Revisiting RREF Form

↳ then the first nonzero number in the row is 1.

↳ if there are any rows that consist entirely of 0's, then they are grouped at the bottom.

↳ If a column contains a leading 1 has 0's everywhere else in that column.

⊗ REF allows for back substitution in solving system of equations.

RREF provides a fully simplified form to read solⁿ directly.

Example:

$$\left. \begin{array}{l} x+y-z=-2 \\ 2x-y+z=5 \\ -x+2y+z=1 \end{array} \right\} \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 2 & -1 & 1 & 5 \\ -1 & 2 & 2 & 1 \end{array} \right]$$

$$R_2 = 2R_1 + R_2$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 3 & 5 & 7 \\ -1 & 2 & 2 & 1 \end{array} \right]$$

$$R_3 = R_1 + R_3$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 3 & 5 & 7 \\ 0 & 3 & 1 & -1 \end{array} \right]$$

$$R_3 = R_3 - R_2$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 3 & 5 & 7 \\ 0 & 0 & -4 & -8 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 3 & 5 & 7 \\ 0 & 0 & -4 & -8 \end{array} \right]$$

→ REF

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & 1 & 5/3 & 7/3 \\ 0 & 0 & 1 & 2 \end{array} \right] \rightarrow$$

$$R_1 = R_1 + R_3$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 5/3 & 7/3 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 5/3 & 7/3 \\ 0 & 0 & 5/3 & 10/3 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array} \right]$$

$\rightarrow$ RREF

$$z = 2$$

$$y + 5/3 z = 7/3 \rightarrow y = \frac{7}{3} - \frac{10}{3} = -1$$

$$x + y - z = -2 \rightarrow x = 1$$

Example:

$$\left[\begin{array}{ccc|c} 1 & 1 & -1 & 7 \\ 1 & -1 & 2 & 3 \\ 2 & 1 & 1 & 9 \end{array} \right]$$

$$R_2 = 2R_2 - R_3 \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & 7 \\ 0 & -3 & 3 & -3 \\ 2 & 1 & 1 & 9 \end{array} \right]$$

$$R_3 = 2R_1 - R_3 \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & 7 \\ 0 & -3 & 3 & -3 \\ 0 & 1 & -3 & 5 \end{array} \right]$$

$$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & 7 \\ 0 & -1 & 1 & -1 \\ 0 & 1 & -3 & 5 \end{array} \right]$$

$$R_3 = R_3 + R_2 \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & 7 \\ 0 & -1 & 1 & -1 \\ 0 & 0 & -2 & 4 \end{array} \right]$$

$$R_1 = R_1 + R_2 \rightarrow \begin{bmatrix} 1 & 1 & 0 & | & 6 \\ 0 & -1 & 1 & | & -1 \\ 0 & 0 & -2 & | & 4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 1 & 0 & | & 6 \\ 0 & -2 & 0 & | & -2 \\ 0 & 0 & -2 & | & 4 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & | & 6 \\ 0 & 1 & 0 & | & 1 \\ 0 & 0 & 1 & | & -2 \end{bmatrix} \quad (RREF)$$